

1. Can you describe your role in relation to the student research projects you have been involved in or supervised?

My involvement in student research projects is primarily content-driven and standards-oriented. As lead of the Critical Raw Materials track, I am responsible for ensuring that student work meets a minimum level of analytical rigor, data quality, and relevance to real-world supply chain challenges.

I do not see my role as primarily facilitative. Rather, I focus on assessing whether students are able to engage seriously with complex material, handle data responsibly, and produce work that can withstand scrutiny beyond the educational setting.

2. At the start of projects, what guidance are students typically given about how they are expected to document their research work? Can you give an example?

Students are informed that their research outputs must be defensible, data-driven, and clearly communicated. They are expected to keep track of their sources, datasets, and analytical steps to a degree that allows their conclusions to be evaluated.

In my experience, this guidance is sufficient. The problem is not a lack of instruction, but rather that students often underestimate the level of discipline required to carry out applied research properly.

3. Can you describe specific situations where documentation came up during project work with students? What happened in those moments?

Documentation issues typically arise when students are unable to justify their conclusions. When asked to explain how a particular insight was derived, it becomes apparent that underlying data is incomplete, poorly structured, or inconsistently referenced.

At that point, documentation is revealed not as a missing administrative step, but as a symptom of insufficient engagement with the data itself.

4. Based on the final outputs of projects, how usable do you think they are for someone new who has not spoken to the original student team? Can you give an example?

Usability varies significantly, but in many cases it is limited. While reports may be written coherently, they often lack the data depth required for meaningful reuse.

In Critical Raw Materials projects, I frequently see descriptive narratives that are not supported by robust datasets. Without reliable data, documentation alone does not make the work reusable.

5. Which parts tend to be difficult for someone new to understand based only on what is written down or stored?

The main difficulty lies in the analytical gaps. It is often unclear how students moved from raw data to conclusions, not because the process was complex, but because it was never fully developed.

This results in work that appears finished on the surface but lacks internal coherence when examined closely.

6. Can you think of an example of an important decision or change during a project, and describe how and where it was documented, if at all?

A recurring issue is that projects change direction due to feasibility constraints, such as limited data availability or time pressure. These changes are usually not documented explicitly, but they are also rarely the result of deliberate methodological reflection.

Instead, they reflect reactive decision-making rather than structured research design.

7. If someone new wanted to understand how one important conclusion of a project was reached, what would they need to look at or read?

They would need access to the datasets used and a clear explanation of how those datasets were analysed. In many cases, this material is either missing or insufficiently developed.

Without solid data foundations, additional documentation does not meaningfully improve understanding.

8. During projects, is it clear who is responsible for documenting what, if anyone? How is that responsibility typically communicated or decided?

Responsibility is assumed to lie with the students. As emerging researchers, they are expected to take ownership of their work, including data handling and documentation.

When this responsibility is not taken seriously, it becomes visible in the quality of the final output.

9. What tools, templates, or systems are students expected to use to document work, and where is everything actually stored in practice?

Students have access to standard academic and collaborative tools. The specific choice of tool is less important than how it is used.

In my view, inconsistencies in storage or tooling are not the primary issue. The core problem is that students do not always use the tools they have with sufficient rigor.

10. What do you think the purpose of documentation is in these projects? Who do you think benefits from it, if anyone? And what, if anything, influences how motivated students are to document their work well?

Documentation serves to support data credibility and analytical transparency. It is not an end in itself.

Students are motivated to document their work properly when they understand that weak documentation reflects weak research. When documentation is treated as a formality, the underlying research quality is usually already compromised.

11. Based on your experience, what tends to get lost most often over time across projects?

What is lost most often is analytical depth. Data is collected, but not interrogated thoroughly. Assumptions remain implicit, and uncertainty is not addressed systematically.

This limits the long-term value of the work, regardless of how well it is written up.

12. How does documentation generally fit into project work from your perspective? Does it seem integrated into the work, or more like something added on top? Why?

Documentation is secondary to data work and analysis. When the research itself is weak, documentation cannot compensate for that.

From my perspective, improving research outcomes requires raising expectations around data competence, not expanding documentation requirements.

13. From your perspective, what currently makes it difficult to achieve consistent documentation across projects?

The main barrier is inconsistent student commitment and skill level. Applied research, particularly in domains such as Critical Raw Materials, demands a level of precision that many students are not yet prepared to deliver.

Without stronger selection, clearer performance expectations, or more rigorous assessment, inconsistency will persist.

14. Is there anything about documentation, continuity, or research practice that we have not discussed but that you think is important?

If we want student research to contribute meaningfully to ongoing work, we need to be honest about its limitations. Not all student output is meant to be reused.

Continuity should be built on high-quality data and serious analytical effort. Without that, documentation becomes an attempt to preserve work that does not merit preservation.